Questions by topic: Water treatment, conservation & new sources - Set 1 (65 marks)

Answer all questions in the spaces provided.

0 1 . 1 Describe **four** factors that should be considered when choosing a site for a dam and reservoir.

[4 marks]

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0 1 . 2 Give **one** advantage and **two** disadvantages of an estuary barrage compared with a reservoir on land.

[3 marks]

Advantage .....

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Disadvantage 1 .....

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Disadvantage 2 .....

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0 1 . 3 Explain why desalination of seawater is only used in some countries.

[2 marks]

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0 1 . 4 Explain how artificial recharge can make aquifer supplies sustainable.

[1 mark]

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0 2 Table 1 shows the effect of installing water meters on the mean daily water use of households in a UK town.

Table 1

	Unmetered households (fixed charge)	**Metered households**
Mean water use / dm^3 per person per day	150	125
Percentage with low-flush toilets	35	58
Percentage using a garden hosepipe weekly in summer	44	20

0 2 . 1 Calculate the percentage reduction in water use per person in metered households.

[1 mark]

Answer ..... %

0 2 . 2 Explain why metering reduces water use.

[2 marks]

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0 2 . 3 Describe **three** ways in which domestic water use can be reduced other than metering.

[3 marks]

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0 2 . 4 Explain why about 20% of the water intended for public supply in the UK never reaches consumers, and how this could be reduced.

[2 marks]

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0 2 . 5 Explain how drip irrigation reduces the water needed to grow a crop compared with spray irrigation.

[2 marks]

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0 3 Figure 1 lists the stages in a water treatment works supplying drinking water from a lowland river.

Figure 1

Stage	Process
1	Screens
2	Sedimentation
3	Aeration
4	Flocculation and clarification
5	Filtration through sand and gravel
6	Activated carbon
7	Sterilisation
8	pH adjustment

0 3 . 1 Explain why clay particles do not settle out in the sedimentation stage, and describe how they are removed in stage 4.

[3 marks]

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0 3 . 2 Explain the purpose of aeration (stage 3).

[2 marks]

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0 3 . 3 Explain why ozone or UV light may be used for sterilisation instead of chlorine when water comes from peaty upland sources.

[2 marks]

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0 3 . 4 Describe how each of the following contaminants is removed from water.

[3 marks]

Toxic metal ions (lead, mercury, arsenic) .....

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Salt .....

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Odours and pesticides .....

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0 4 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 4 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 4 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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0 5 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

0 5 . 2 Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

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0 5 . 3 Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

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0 6 Table 1 shows the water quality required for four different uses.

Do not write outside the box

0 6 . 1 Complete Table 1 by giving the quality requirement for each water use and the reason for it.

[4 marks]

Table 1

Water use	Quality requirement	Reason
Potable (public) supply		
Spray irrigation		
Power station condenser cooling		
Industrial boiler water		

0 6 . 2 Explain why aeration is used in the treatment of water from anaerobic sources.

[2 marks]

0 6 . 3 Explain how flocculation removes clay particles that sedimentation cannot.

[2 marks]

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0 6 . 4 Give **two** methods used to sterilise treated water.

[2 marks]

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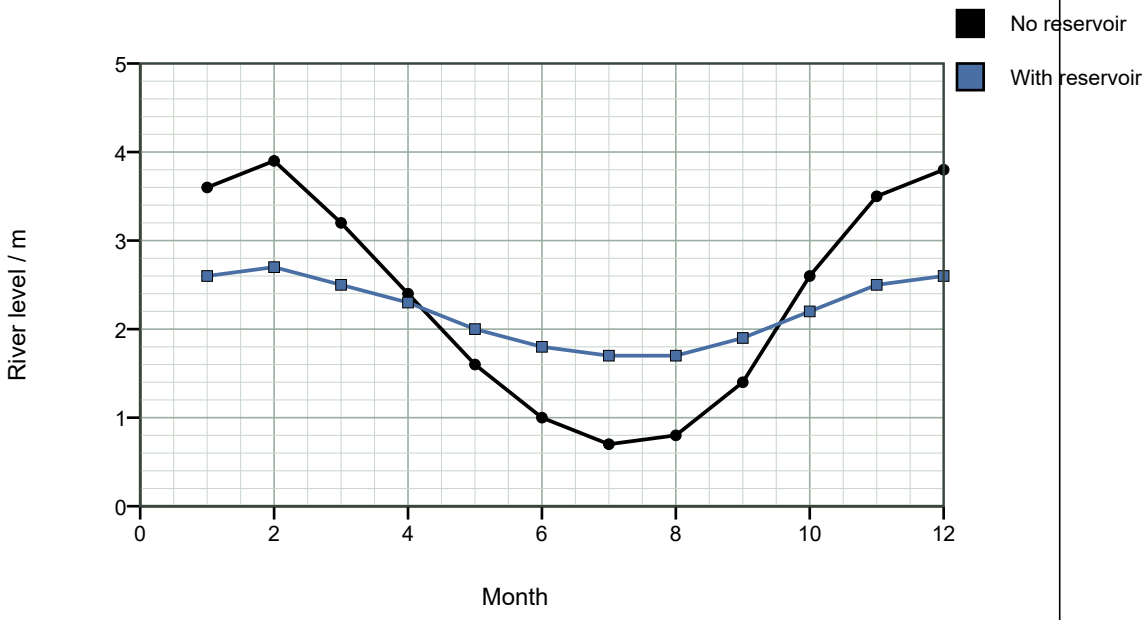
2 .....

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0 7 Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



0 7 . 1 Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer ..... m

0 7 . 2 Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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Do not write outside the box

0 7 . 3 Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

0 7 . 4 Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

15

END OF QUESTIONS